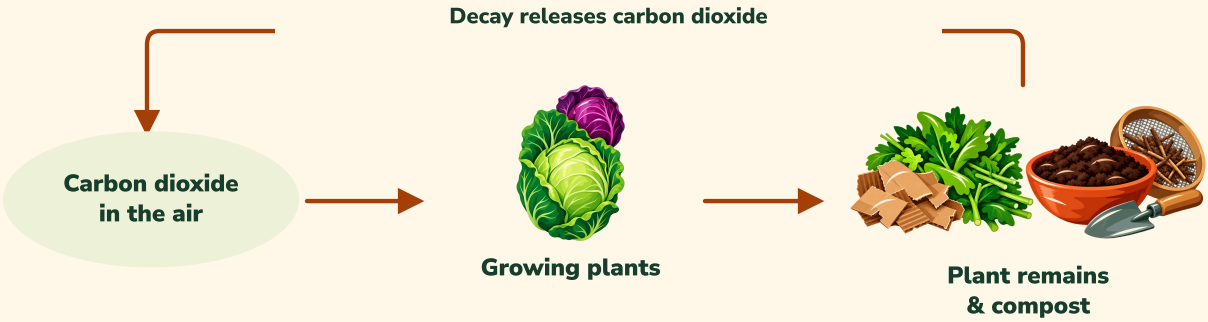
Nutrient cycles Feed the soil

Last year's plant material can help grow this year's crops. Compost returns useful material to the ground, and soil organisms make it available again: a practical cycle at the heart of organic growing.

Carbon: from air to soil

Plants use sunlight to turn carbon dioxide from the air into leaves, stems and roots. When this material decays, fungi and bacteria feed on it, releasing nutrients and returning much of the carbon to the air.



Humus: the part that lasts

Some organic matter breaks down slowly, forming the darker, longer-lasting material we call humus. It helps soil hold water and nutrients and form crumbs that roots can grow through. Mature compost supplies organic matter as well as plant food.

Nitrogen: a helping hand from legumes

Most plants cannot use nitrogen straight from the air. Peas and beans have bacteria in root nodules that can fix it into a form the plant can use. As healthy roots and other remains decay, some becomes available to later crops.

Cut spent peas and beans at ground level and leave healthy roots in place; compost the tops. Follow with leafy crops, as in the rotation example. This contributes to their feed; add mature compost where needed.

Nitrogen in the air → **Root-nodule bacteria** → **Peas & beans** → **Decay feeds later crops**

Phosphorus & potassium: recycle these too

Phosphorus · roots and growth

Phosphorus supports roots and energy transfer. Compost returns it to the soil; acidity and soil conditions affect how much plants can use.

Potassium · fruit and water

Potassium helps plants manage water and develop fruit. Compost and comfrey feed recycle it; comfrey gathers potassium already in the soil.

John: Why We Make Compost; pea/bean JSON. Soil cycles: RHS, Garden Organic, AHDB.